Modern Physics SCP 233

Optics and Special Relativity Lab Report

Esteban Salcedo*[a,1]

[a] *LaGuardia Community College, CUNY*

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Abstract—This laboratory report focuses on exploring the experiments and concepts that opened the door to modern physics. It introduces key ideas and experiments that challenged classical intuition, from the double-slit experiment and optical interference to the invariance of the speed of light and the development of special relativity, which addressed limitations in classical mechanics.
The aim of this work is to reproduce and analyze some of these experiments using laboratory equipment and simulations, recording the corresponding data and gaining insights by comparing experimental results with theoretical predictions. Through this approach, the relationship between theory and observation is emphasized, showing how modern physics emerges from experimental evidence.
In addition, two experimental proposals are presented. The first is a conceptual recreation of the Michelson–Morley experiment using a rotational interferometer setup. The second is an experiment designed to test time dilation using laser pulse measurements to track distances to the International Space Station, combined with atomic clocks for precise timing, and reflective panels similar to those used in lunar laser ranging. Overall, this work connects optical phenomena and relativistic effects within a unified experimental framework, illustrating how fundamental physics can be explored through both simulations and real-world experimental designs

Keywords—*Young's double-slit experiment, interference, diffraction, Michelson interferometer, Michelson–Morley experiment, special relativity, special relativity, simultaneity, optical path difference, fringe pattern*

1. INTRODUCTION

1.1. Interference and Young's Experiment

Interference is a wave phenomenon that occurs when two sources of waves come together and interact with each other. It refers to the situation in which two waves pass through the same region of space at the same time [2]. A good example is producing pulses on a cord attached to a wall with the hand; the pulses reflect from the wall and recombine with new pulses coming from the hand.

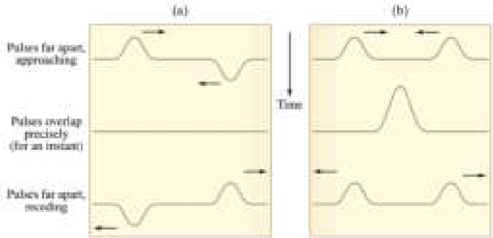

Figure 1. Two wave pulses overlapping, showing (a) destructive interference and (b) constructive interference [2].

Light, once thought to be only a particle by Isaac Newton, also exhibits wave behavior, as proposed by Christiaan Huygens. This dual nature was later demonstrated in Young's double-slit experiment, which provided strong evidence of the wave properties of light through the observation of interference patterns [3].

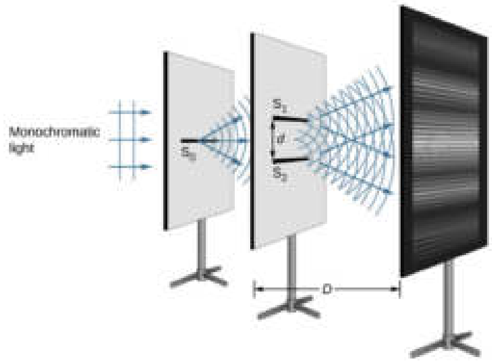

Figure 2. The double-slit interference experiment using monochromatic light and narrow slits. [3].

The experiment consisted of sunlight passing through a pinhole on a first screen, then through two pinholes on a second screen, and eventually being projected onto a surface where the interference pattern was observed [3].The screen displayed wave behavior, with regions of high intensity and others with little or no intensity as shown in Figure 2. By varying the distance between the slits on the second screen, patterns of constructive and destructive interference were produced.

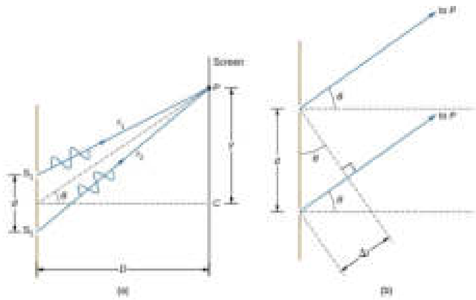

Figure 3. (a) To reach P, the light waves from S1 and S2 must be different, (b) the path difference between the two rays is Δt [3].

This variation between constructive and destructive interference does not occur directly because the slit separation d is changed, but because changing d alters the optical path difference Δt between the two light beams.

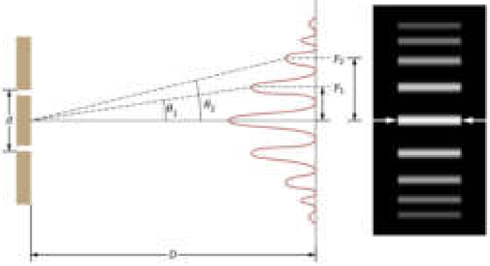

Figure 4. The interference pattern for a double slit has an intensity that falls off with angle. The image shows multiple bright and dark lines, or fringes, formed by light passing through a double slit [3].

*esteban.salcedo@live.lagcc.cuny.edu

Interference, Diffraction, and Relativistic Effects in Modern Physics

The following relation describes the path difference responsible for the interference taking place.

$$d \sin\theta = m\lambda \tag{1}$$

$m = 0, \pm1, \pm2, \pm3\ldots$ represents the orders of the fringes. If the light source remains constant (that is, the wavelength λ does not change) and a fixed fringe m is considered, then a smaller slit separation d results in a larger angle θ, since $sin\theta = m\lambda/2$. For small angles, the approximation $\sin\theta \approx 0$ can be used.

$$y = D\tan\theta \approx \frac{m\lambda D}{d} \tag{2}$$

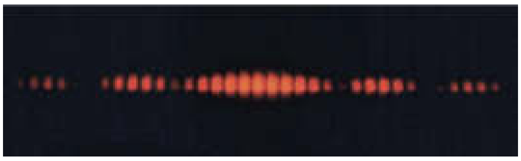

Figure 5. Fringes obtained from the double slit experiment [**Intelaria**].

In addition to interference, a new concept must be introduced: diffraction. It plays an important role in the observed pattern. Diffraction occurs when light passes through a finite-width slit and spreads out, producing an intensity envelope that limits the visibility of the interference fringes.

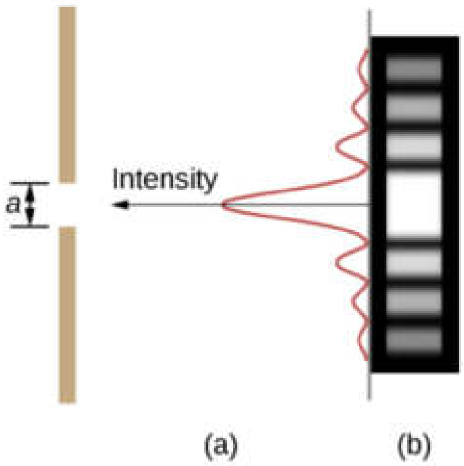

Figure 6. Light passing through a single slit producing a diffraction pattern [3].

This effect is described by the single-slit condition:

$$a \sin\theta = m'\lambda \tag{3}$$

Where a is the slit width. This condition determines the positions of the minima in the diffraction pattern and explains why the intensity of the double-slit fringes decreases away from the center. Equation 3 looks very similar to equation 1. However they are both representing different behaviors of light.

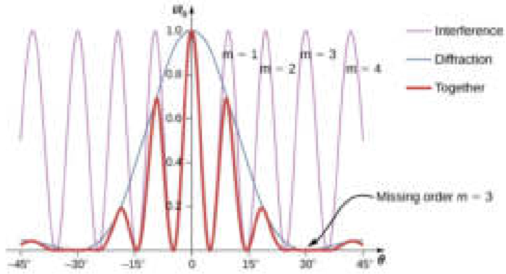

Figure 7. Diffraction and interference effects occurring in the same setting. Diffraction reduces the overall intensity of the pattern, while interference determines the number of visible fringes. [3].

A parameter of interest for these combined effects is the first minimum, where the intensity decays significantly. The angle corresponding to a given order m in the interference pattern is the same as the angle of the first minimum m' in the diffraction pattern.

$$sin\theta = \frac{m'\lambda}{a} = \frac{m\lambda}{d} \tag{4}$$

From this the number of bright fringes before the intensity decays is given by 5

$$m = \frac{d}{a} \tag{5}$$

The fringes are symmetrically arranged, with one located at the center corresponding to $m = 0$, where the intensity is maximum. This pattern can be reproduced in a laboratory setting using appropriate measurement instrumentation. This experiment is considered one of the most important in physics, as it opened the door to modern physics and later the development of quantum mechanics.

1.2. Michelson Interferometer

The Michelson interferometer was invented by the American physicist Albert A. Michelson. It uses the principle of light interference by splitting a light beam into two paths and recombining them after they travel different optical distances, ultimately producing an interference pattern of fringes on a screen[3].

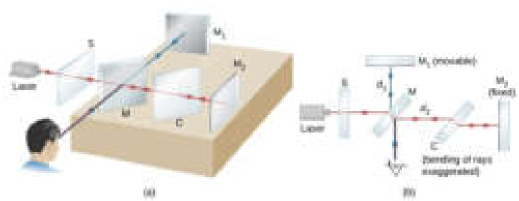

Figure 8. (a) Michelson interferometer showing the splitting of a light beam and its recombination after reflection from the mirrors. (b) A planar view of the interferometer [3].

Similar to Young's experiment, a coherent light beam of wavelength and intensity I_0 is split into two beams of lower intensity. Each beam travels along a separate path, with one directed toward mirror 2, which becomes the reference path. The position of this mirror can be adjusted with high precision, allowing the optical path length to be varied by micrometers or even nanometers. This makes the Michelson interferometer a highly sensitive and precise measurement system.

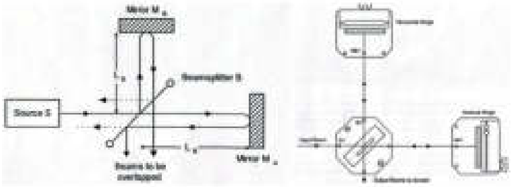

Figure 9. A simplified Michelson interferometer [9].

The advantage of this model for producing light interference, compared with the one designed by Thomas Young, is the ease of varying the phase between the two beams. There is not longer a double slit feature altering producing a pathway t by varying the slits with a bigger distance d, but instead, we can adjust t directly by changing distance L_B by moving the mirror M_A as shown in Figure 9. The interference patter is gonna be describe as the equation 6 for constructive interference and equation 7 for destructive interference:

$$2\Delta t = m\lambda \tag{6}$$

$$2\Delta t = (m + \frac{1}{2})\lambda \tag{7}$$

1.3. Michelson-Morley Experiment

In the nineteenth century, scientists believed that waves necessarily required a medium to propagate. Just as water waves require water, a vibrating string requires the string itself, and sound waves require air to transfer energy, it was assumed that light also needed a medium through which to travel [1].

Michelson realized that the effect of Earth's motion on direct measurements of the speed of light would be too small to detect. Instead, he proposed using interference as a more sensitive method by comparing differences in light travel time along different paths. This led to the development of the Michelson interferometer. The goal of the Michelson–Morley experiment was to measure the speed of light relative to Earth in order to detect its motion through the supposed ether and confirm its existence.

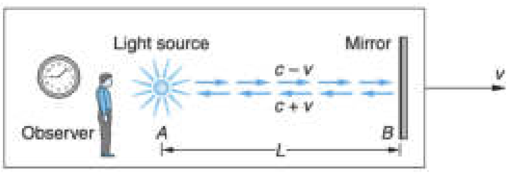

Figure 10. Observer, light and mirror traveling with a speed v relative to the weather [1].

According to classical theory, the speed of light relative to the ether would appear different to an observer moving with velocity v. For light traveling in the direction of motion, the speed would be $c - v$, while for light reflecting back toward the source, it would be $c + v$. Before discussing the experiment carried out by Michelson, an analogous example can be used to better illustrate the underlying idea of the experiment.

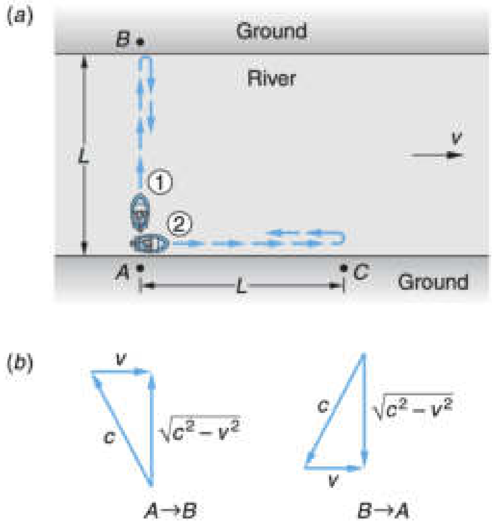

Figure 11. (a) Two rowers, each moving with speed c relative to the water, travel the same distance L. A current with speed v flows along the path of rower 2. (b) Rower 1 must aim upstream so that the combination of their velocity and the current results in a straight path from A to B and back from B to A [1].

In Figure 8, two rowers travel the same distance L at the same speed c, but along perpendicular directions. If there were no current acting as a disturbance, both rowers would arrive at the same time according to classical physics. However, because a current is present, one of the rowers experiences a delay, resulting in different travel times. So, the question is who is going to win?

Let us analyze the time for rower 1 from A to B and from B to A:

$$t_1 = \frac{2L}{\sqrt{c^2 - v2}} = \frac{2L}{c}(1 - \frac{v^2}{2c^2})^{-\frac{1}{2}} \tag{8}$$

Using the binomial expansion theorem

$$t_1 \approx \frac{2L}{c}(1 + \frac{v^2}{2c^2} + ...) \tag{9}$$

Now for rower 2 from A to C and C to A:

$$t_2 = \frac{L}{c+v} + \frac{L}{c-v} = \frac{2Lc}{c^2 - v^2} \tag{10}$$

$$t_2 = \frac{2L}{c}(1 - \frac{v^2}{c^2})^{-1} \tag{11}$$

Applying the binomial expansion theorem

$$t_2 \approx \frac{2L}{c}(1 + \frac{v^2}{c^2} + ...) \tag{12}$$

If t_2 is grater than t_1, then

$$\Delta t = t_2 - t_1 > 0 \tag{13}$$

$$\Delta t = \frac{2L}{c}(1 + \frac{v^2}{c^2}) - \frac{2L}{c}(1 + \frac{v^2}{2c^2}) = \frac{Lv^2}{c^3} \tag{14}$$

Since Δt is higher than cero, t_2 is bigger than t_1 so rower 1 is going to win the race.

With that analogy in mind, the two rowers can be interpreted as two light beams affected by a hypothetical "ether wind" flowing along one of the paths. This would introduce a time delay between the beams, which would ultimately lead to a detectable interference pattern. That was what Michelson and Morley had in mind when carrying out the experiment, creating a much-improved interferometer with path L being about 11 meters as a result of multiple reflexions.

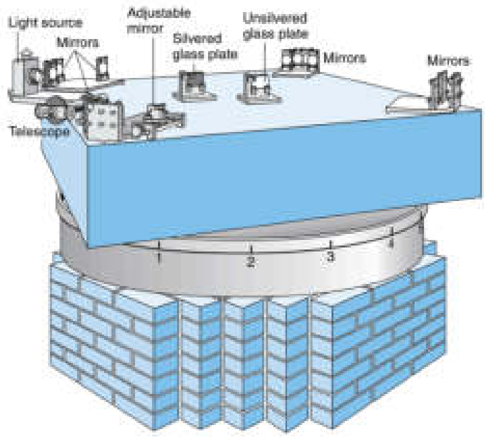

Figure 12. Schematic of the Michelson–Morley interferometer mounted on a 5 ft square sandstone slab floating on mercury, designed to reduce vibrations during rotation [1].

Michelson and Morley's understanding of interference did not depend only on the number of waves in each path, but also on the speed of the light relative to the observer. The idea of light having different speeds was based on a Newtonian view of relativity. According to this, rotating the interferometer by 90 degrees would change the time difference between the two paths, alter the phase, and produce a shift in the interference fringes[1]. However, what was predicted to happen did not occur, and it became a great disappointment for Michelson, Morley, and much of the scientific community [1]. Many later attempts were made to recreate the experiment with more precise setups, but the same result was obtained. The shift in the fringes did not occur, and the observed change was only about 0.01% [1] for Michelson and Morley, which was within the experimental uncertainty of the apparatus.

From this experiment, a small but important gap was revealed where Newtonian physics was no longer sufficient. The invariance of the speed of light between inertial reference frames became like a new "apple falling," but this time it was Albert Einstein who recognized its significance and developed a new understanding of physics.

1.4. Special Relativity

Albert Einstein published his revolutionary theory of special relativity in 1905, opening the door to a new way of understanding the world. His theory is founded in two postulates:

- The laws of physics are the same in all inertial frames.
- The speed of light in vacuum is equal to the value c, independent of the motion of the source.

The first postulate implies that there is no preferred reference frame, or one more valid than another; therefore, absolute motion cannot be detected. The second postulate states that the speed of light is the same for all observers, independent of the motion of the source or the observer. This can be compared to other waves, such as sound waves, whose speed in a given medium remains constant even if the source is moving, like a car. Although the Doppler effect is produced—changing the frequency detected by the observer—the wave itself continues to travel at the same speed in that medium [1]. What reinforces this postulate is that no medium has been found through which light propagates, as discussed previously. According to this, all observers measure the same value c for the speed of light, regardless of their motion. These two postulates sound reasonable, yet when considered together, they seem to break the common sense of the world as it has been known for years.

One of the concepts that changes is simultaneity, which is no longer absolute. Albert Einstein illustrated this with a thought experiment involving a fast-moving train. An observer stands on a platform outside the train, while another observer is seated at the exact center inside the train, moving along with it. As the train passes the platform, two lightning bolts strike the front and the back of the train. From the perspective of the observer on the platform, the two strikes occur at the same time, and since the light from both strikes reaches him simultaneously, he concludes that the events were simultaneous. However, for the observer inside the train, the situation appears different. Because he is moving forward with the train, he encounters the light from the front strike before the light from the back strikes. As a result, he concludes that the lightning strike at the front happened before the one at the back. This example shows that two events that are simultaneous for one observer may not be simultaneous for another, depending on their relative motion [3].

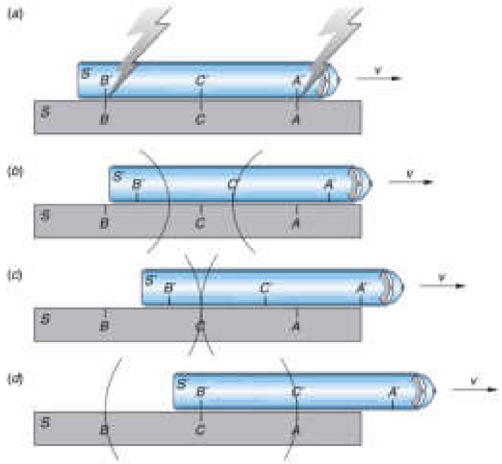

Figure 13. (a) Lightning bolts strike both the front and the back of the train. (b) For observer C, who is standing on the platform, the strikes are simultaneous. (c) Observer C', inside the train, first experiences the flash from the front of the train. (d) Observer C' later experiences the flash from the back of the train [1].

Another concept that changed as a result of these two postulates is time. Once again, a thought experiment is used to analyze this. A traveler is inside a vehicle moving at a very high constant velocity (with no acceleration). Inside, he measures a pulse of light traveling vertically a distance D, reflecting off a mirror, and returning over a time interval $\Delta\tau$. An observer outside the ship watches the same pulse of light, but from his perspective, the light follows a longer, diagonal path due to the motion of the vehicle over a time interval Δt (Figure 14). Since the speed of light remains constant for both observers, we conclude that the time measured by the observer outside the ship is greater than the time measured by the observer inside the ship [3].

Time dilation refers to the increase in the measured time between two events when observed from a frame that is moving relative to the frame where the events occur at the same position. Let us use the equations in figure 14 (c) to obtain the relationship between the two times.

$$S = \sqrt{D^2 + L^2} \tag{15}$$

Knowing that $D = c\Delta\tau/2$ and $S = c\Delta t/2$, then

$$(\frac{c\Delta t}{2})^2 = (\frac{c\Delta\tau}{2})^2 + (\frac{v\Delta t}{2})^2 \tag{16}$$

$$\Delta\tau^2 = \Delta t^2 - \frac{v^2\Delta t^2}{c^2} \tag{17}$$

$$\Delta\tau^2 = \Delta t^2(1 - \frac{v^2}{c^2}) \tag{18}$$

$$\Delta\tau = \Delta t\sqrt{1 - \frac{v^2}{c^2}} \tag{19}$$

$$\Delta t = \frac{\Delta \tau}{\sqrt{1 - \frac{v^2}{c^2}}} \tag{20}$$

This result is equivalent to $\Delta t = \gamma \Delta \tau$, where γ is known as the Lorentz factor ($\gamma \geq 1$).

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \tag{21}$$

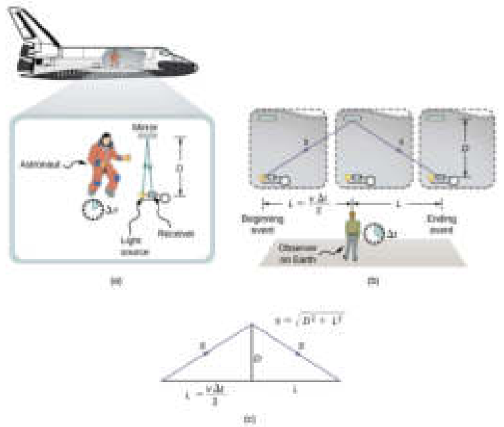

Figure 14. (a) An observer inside the ship measures time for light to travel $2D$. (b) An observer outside measures time for the same light pulse to travel S (c) This triangles help us obtain the distance s with Pythagoras theorem [3].

According to equation 20, the observer outside the ship is experiencing a larger time Δt than the observer inside the ship, who measures a smaller time $\Delta \tau$

The proper time interval is the time measured by an observer who sees both events taking place at the same position in space. If an observer is located where both events happen at the exact same position (for example, inside the moving ship with the light clock), then the time they measure between those events is the proper time.

The Lorentz transformation is a fundamental concept in special relativity that describes how measurements of space and time change between different inertial reference frames moving at constant velocity relative to each other. Unlike classical mechanics, where time and space are treated as absolute, the Lorentz transformation shows that they are interconnected and depend on the observer's motion. This transformation is essential for understanding relativistic effects such as time dilation and length contraction, which become significant when objects move at speeds close to the speed of light.

$$t' = \gamma \left(t - \frac{vx}{c^2} \right) \tag{22}$$

$$x' = \gamma (x - vt) \tag{23}$$

Where equation 22 corresponds to the time transformation and equation 23 corresponds to the position transformation.

1.5. Overview of Experimental and Simulated Approaches in Modern Physics

The goal of this paper is to reproduce some of the experiments carried out throughout the years that opened the door to what is called modern physics. It begins with the production of interference patterns using single- and double-slit setups, aiming to record a clear interference pattern. It also includes simulations of a Michelson interferometer with the objective of calculating the wavelength of a given light source.

In addition, a setup is proposed that would allow the Michelson–Morley experiment to be conceptually reproduced as it was originally performed. Furthermore, a simulation of time dilation is used, where an outside observer provides data of time intervals in a moving vehicle in order to calculate its velocity.

Finally, a proposal for a real and well-detailed experiment for time dilation is presented, explaining the setup and experimental conditions.

2. METHODS

2.1. Double-slit Experiment

An optical setup was used to investigate interference and diffraction patterns produced by a double-slit pattern. A coherent monochromatic light source was directed toward interchangeable slit plates. The set up is equipped with the following parts:

- Optic bench
- Rail bench for the sensor
- Optical intensity sensor
- Data acquisition unit (Vernier interface)
- Interchangeable slit set
- Red diode laser set (635 nm wavelength)

Figure 15. Laboratory Double-Slit Experiment Equipment

The overall assembly was aligned along a common optical axis to ensure consistent beam propagation and accurate detection of fringe patterns Figure 15.

In order to obtain a well-defined fringe pattern, different parameters were varied. The variables that were changed included the number of slits, the screen distance, the sensor sensitivity, and the slit width.

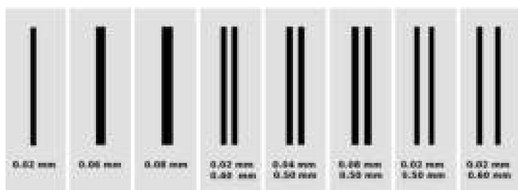

Figure 16. Double Slits with Different Widths and Separation Distances

The interchangeable slit set included several options, such as single slits, double slits, multiple slits, and combined configurations. A setup was selected and tested while observing the screen. If the result was not satisfactory, the distance or slit width was adjusted to produce different interference patterns, and the setup was always inspected to ensure there was no dirt or obstruction, as this could affect the interference pattern.

Figure 17. Varying screen distances to manipulate the fringes

The screen distance (Figure 17) was a parameter that played a major role in producing an interference pattern on the screen. The slit setup was moved along the optical bench while recording the position where the pattern was most clearly observed.

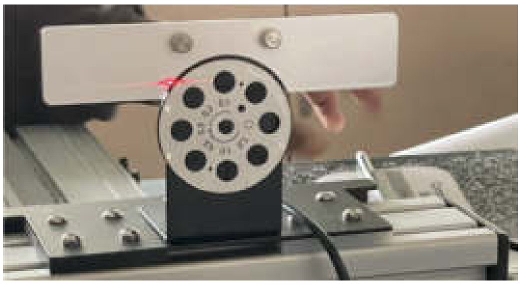

Figure 18. Sensor capturing light intensity while moving side to side

The optical sensor had three sensitivity levels, which controlled the amplification of the detected light signal sent to the Vernier interface. During measurements, the sensor was moved slowly from one side to the other along its rail. This allowed the sensor to collect more data, resulting in a smoother and more accurate final intensity graph.

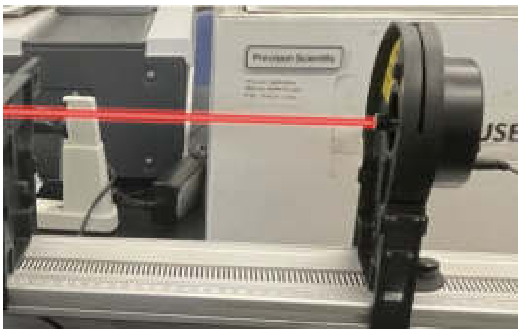

Figure 19. Diode laser with adjustable alignment and intensity thumb screws at the back

Adjusting the laser is necessary to control the intensity of the emitted light beam and to align its direction. Thumbscrews located at the back of the device are used for these adjustments. It is always important to avoid direct exposure of the eyes to the laser beam, as it can cause permanent eye damage.

Figure 20. Vernier interface data collector creating intensity–displacement curves [**Vernier**]

The data collected by the sensor was plotted and recorded in the Vernier interface. Once a clear graph was obtained, it was saved along with other graphs taken under different parameters in order to compare and analyze the results.

2.2. Michelson Interferometer Simulations

A simulation of the Michelson interferometer was used to perform this part of the experiment. The simulation was obtained from the Javalab website [6].

Figure 21. Michelson interferometer simulations with visible light waves creating constructive and destructive interference [6]

This simulation offered several adjustable features, such as the frequency of the light, the distance of mirror 1, freezing the simulation, and hiding one of the rays. At the start, the simulation automatically emitted a light beam, which was split upon reaching the beam splitter in the center. The two resulting beams traveled toward mirrors 1 and 2, where they were reflected back. Upon returning to the beam splitter, the beams were partially recombined and redirected, eventually producing an interference pattern.

For the experiment, a specific wavelength was selected and constructive interference was produced. The corresponding positions were recorded using the ruler provided in the simulation. The mirror was then moved to reach the next constructive interference condition, and the new position was recorded. From these measurements, the optical path difference was obtained. Equation 6 was then used to determine the wavelength of the light source.

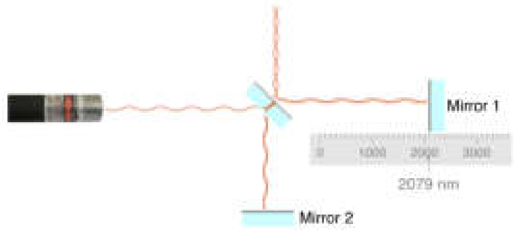

Figure 22. Destructive interference created as a result of moving the mirror by a distance Δd. [6]

This simulation illustrated the wave behavior of the system along the optical paths; however, it did not include a screen for directly observing the fringes or interference pattern. Instead, it served as a simple and accessible tool for developing an initial understanding of the concept.

A second resource for observing fringe patterns was provided by the University of Ceará [10] simulation website (Figure 23).

Figure 23. Michelson interferometer simulation with a fringe screen and a cuvette containing gas placed along the light path [10]

The simulation was operated by first turning the laser on and selecting the desired color. The option "View Light Trajectory" was activated to observe the propagation of the beam. A cuvette was then placed or removed using the corresponding control button to introduce different types of gas along the optical path. The micrometer was activated by pressing the small yellow button, allowing fine adjustments of the system. Finally, the effects of interference were observed as changes in the path difference and refractive index produced variations in the resulting pattern.

2.3. Michelson–Morley Proposal

The goal of this section of the lab was to propose a version of the Michelson–Morley interferometer in order to obtain similar results, showing that there is no detectable medium through which light travels, as explained previously.

Figure 24. Michelson interferometer laboratory equipment setup [9]

It is proposed to design a base or rolling support using a square rotating bearing plate, secured with four screws to a rigid base and another four screws to the setup of the current Michelson interferometer.

Figure 25. Square rotating bearing plate [**The Home Depot**]

The Michelson interferometer setup is mounted on a rotating platform, giving it an additional degree of freedom. Cables must be secured to the board to prevent them from moving freely. It is recommended to cover the edges of the interferometer board (like a fence) for added protection.

2.4. Relativity Experiment

A simulation from the JavaLab website [7] was used to study the effect of relative motion on time measurements. The simulation represents a train moving at high speed with a light source located at its center and two sensors positioned at the ends.

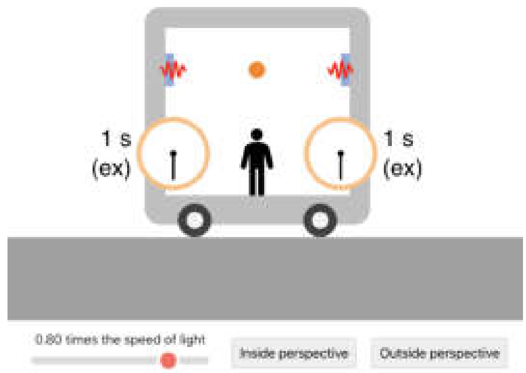

Figure 26. A train traveling at speed v with a light source at the center, two sensors at the ends, and observers inside and outside [7].

The experiment consists of selecting six different points in which the observer inside the vehicle measures a time interval of 1 s for a light pulse to reach each of the sensors, while the external observer records different time intervals for each case. Using the Lorentz transformations, the velocity of the vehicle is calculated for each configuration.

2.5. Proposal for a Time Dilation Experiment

The ISS (International Space Station) orbits the Earth at approximately 5 miles per second (about 8 km/s) [8]. This is a very high speed that remains nearly constant, which is convenient for the experiment.

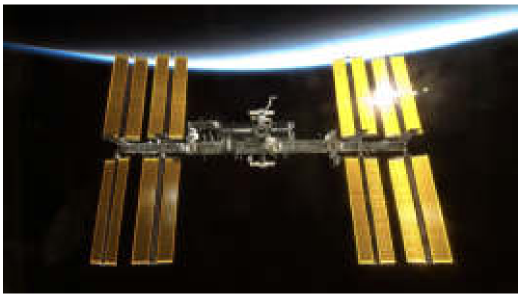

Figure 27. The ISS orbiting the Earth at approximately 8 km/s [8]

The ISS is equipped with an atomic clock provided by the ESA (European Space Agency), with the purpose of creating a "network of clocks" with another atomic clock on Earth, and comparing them in order to measure the flow of time [5].

Figure 28. ACES: an atomic clock launched on April 21st, 2025 by the ESA [5]

The experiment compares time measurements between an observer on Earth and an observer on the ISS using light signal exchange and atomic clocks. The necessary equipment to carry out the experiment is:

- Atomic clocks (Earth station and ISS)
- Stabilized laser source
- Laser reflecting panels (corner cubes)[4]
- Photodetector
- Data acquisition system
- Communication interface between measurements

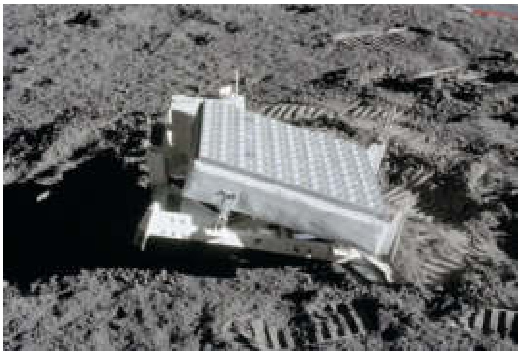

Figure 29. A laser reflecting panel deployed on the Moon by Apollo 14 in 1971 [4]

This panel in Figure 29 is made of 100 mirrors that reflect light back in the same direction it comes from [4]. For the experiment, one of these must be positioned on the ISS and the other on Earth.

Figure 30. Laser-ranging facility at the Goddard Geophysical and Astronomical Observatory in Greenbelt, Md [4]

Atomic clocks at both the Earth station and the ISS, together with a stabilized laser source, laser reflecting panels, photodetectors, and a data acquisition system, are used to carry out the experiment, while a communication interface allows synchronization between measurements.

A laser pulse is first sent from the Earth station toward the ISS (Figure 28). The signal is reflected back by the laser reflecting panels and the Earth station records the emission and return times using its atomic clock to determine the round-trip time. The ISS simultaneously records the reception and re-emission times using its onboard atomic clock and photodetector. The same process is then repeated in the opposite direction, with the ISS sending a laser pulse to Earth and the Earth station reflecting it back.

Both observers compare their measured time intervals for the same exchange of signals. The ISS is assumed to move at a constant orbital speed of about 8 km/s during the experiment, and atmospheric effects are neglected or corrected. Finally, the recorded times from both reference frames are compared to analyze any differences related to relativistic effects.

3. RESULTS

3.1. Experimental Results and Analysis of Diffraction and Interference Patterns

Table 1. Settings for different fringe patterns using in a single-slit experiment

Setting	Sensor Scale	Screen Distance (cm)	Slit Width (mm)	Angle 1st Min (°)
1	1	69.4	0.02	1.819
2	1	69.4	0.04	0.909

Table 2. Settings for different fringe patterns using in a double-slit experiment

Setting	Sensor Scale	Screen Distance (cm)	Separation d (mm)	Slit Width (mm)
1	1	89	0.5	0.04
2	10	90	0.5	0.04
3	10	92	0.5	0.04
4	1	87	0.5	0.08
5	1	90	0.5	0.08
6	1	95	0.5	0.08

Table 3. Theoretical Values for First Minimum and Missing Order for Double-slit experiment

#	Angle 1st Min (Theory) [°]	Missing Order (Theory)
1	0.909	13
2	0.909	13
3	0.909	13
4	0.454	7
5	0.454	7
6	0.454	7

Table 4. Experimental Values for First and Missing order for Double-slit experiment

#	Angle 1st Min (Exp) [°]	Missing Order (Exp)
1	0.800	11
2	0.209	5
3	0.800	11
4	0.436	6
5	0.509	7
6	0.430	6

Table 5. Errors in Angles and Missing Order for the Double-Slit Experiment

#	Angle Error [%]	Missing Order Error [%]
1	12.1	15.3
2	77	61.5
3	12.1	15.3
4	3.96	14.28
5	12.11	0
6	5.6	14

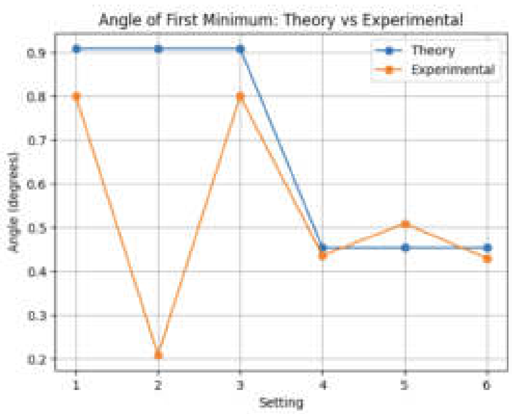

Figure 31. First Minimum Angle: Theory vs Experimental

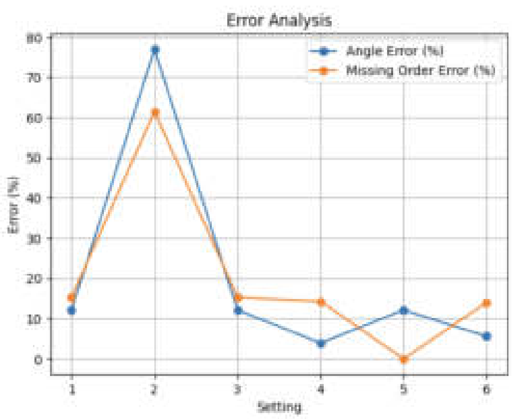

Figure 32. Error Analysis

3.2. Fringes due to interference and diffraction

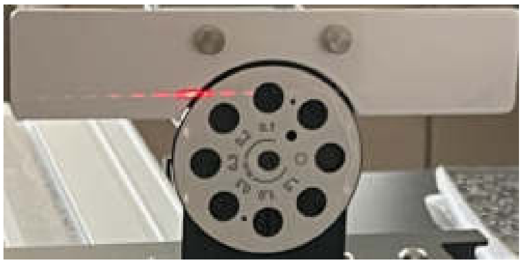

Figure 33. Photography of fringes in Setting 2 for double-slit interference

Figure 34. Photograph of fringes in Setting 2 for single-slit interference

3.3. Graphs Recorded for Single-slit t Interference

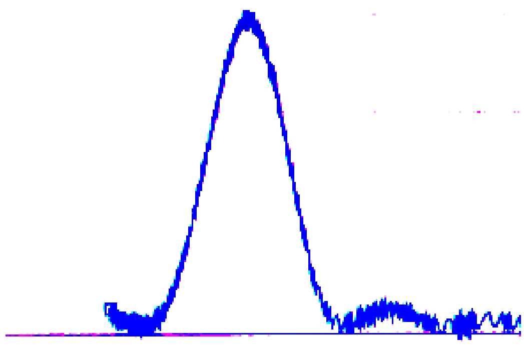

Figure 35. Single-slit fringe graph for the first setting

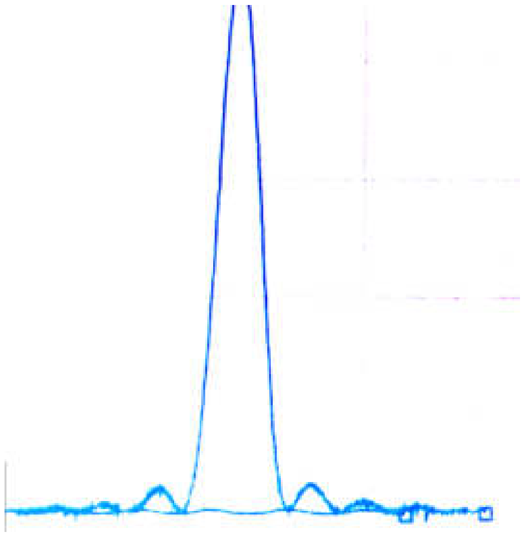

Figure 36. Single-slit fringe graph for the second setting

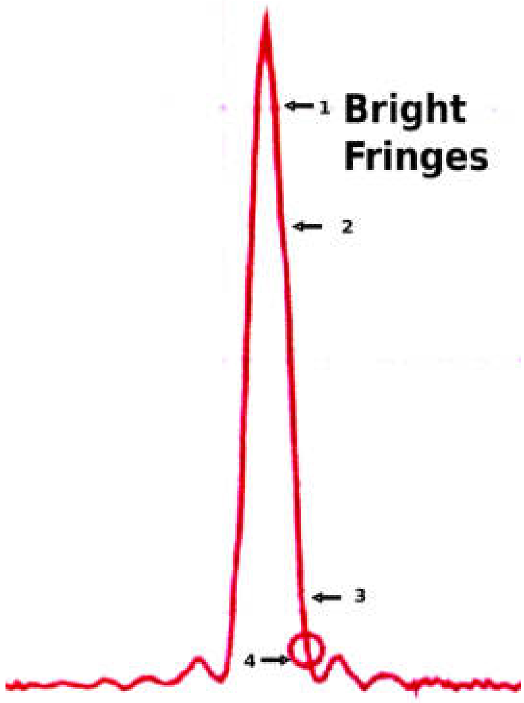

Figure 38. Interference Pattern in the Second Double-Slit Configuration

3.4. Graph Recorded for the Double-Slit Experiment Tracking Bright Fringes Before the First Minimum

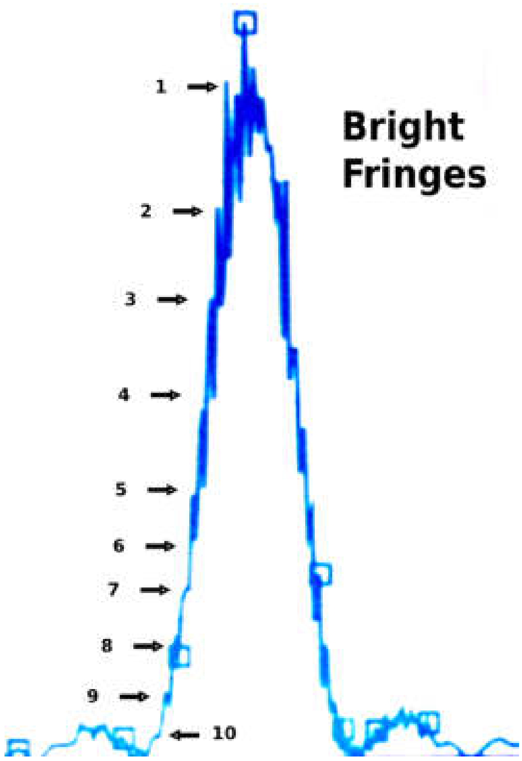

Figure 37. Interference Pattern in the First Double-Slit Configuration

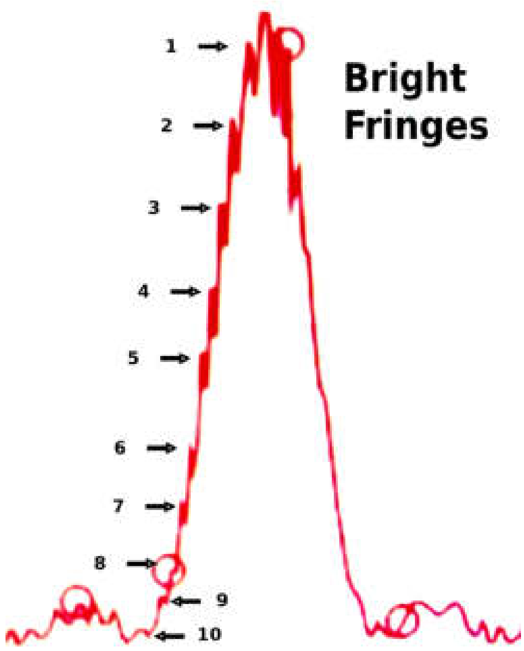

Figure 39. Interference Pattern in the Third Double-Slit Configuration

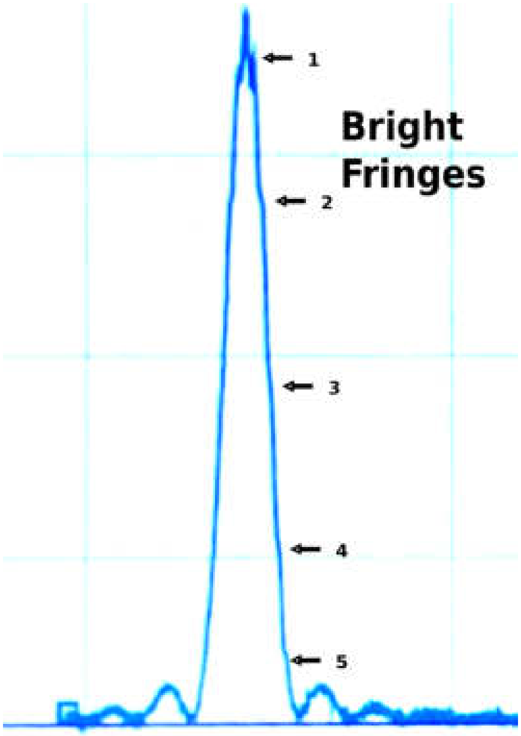

Figure 40. Interference Pattern in the Fourth Double-Slit Configuration

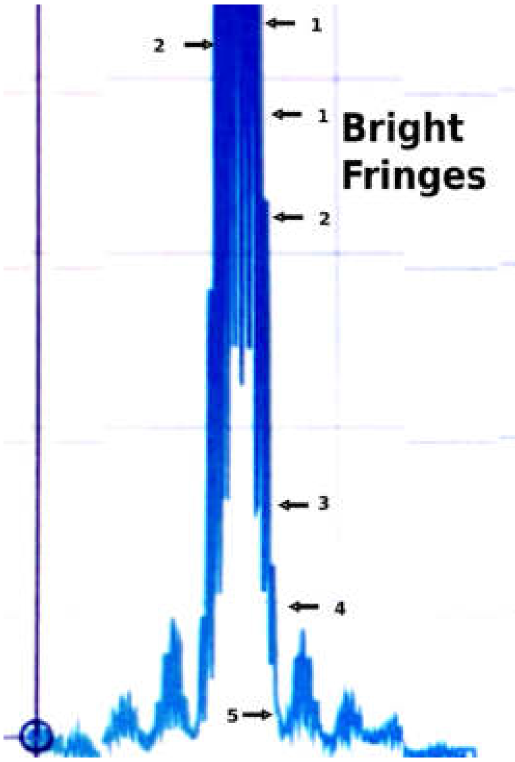

Figure 42. Interference Pattern in the Sixth Double-Slit Configuration

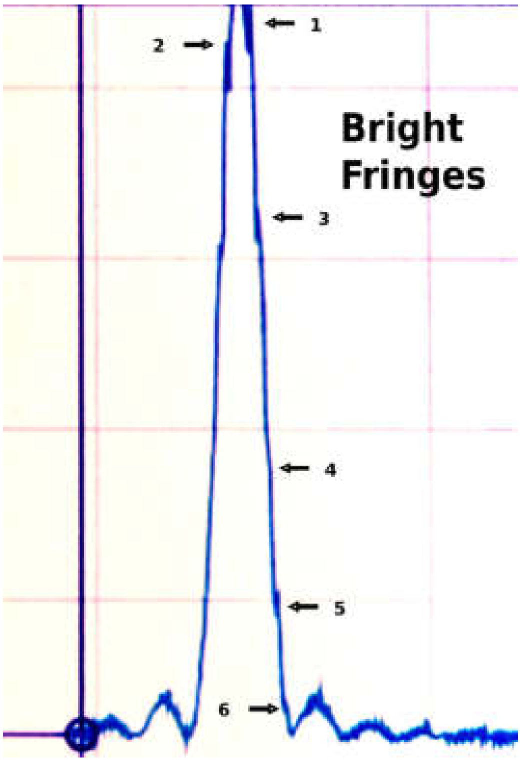

Figure 41. Interference Pattern in the Fifth Double-Slit Configuration

3.5. Contrast Between Single-Slit and Double-Slit Patterns

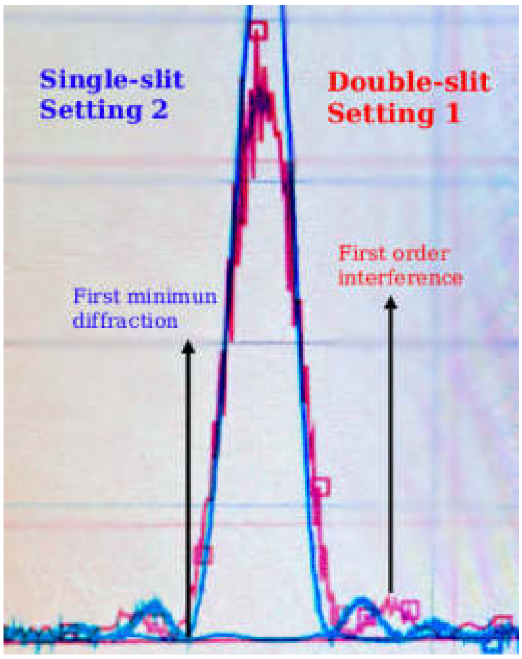

Figure 43. Single-slit setting 2 and double-slit setting 1

3.6. Calculating wavelength for different light sources with simulation[6]

Table 6. Measured positions and optical path difference in a Michelson interferometer

Point	Position 1 (nm)	Position 2 (nm)	Δt (nm)
1	1316	1014	302
2	1041	757	284
3	1325	1058	267
4	1076	845	231
5	1316	1094	222
6	1316	1103	213

Table 7. Experimental and nominal wavelength with percentage error

Point	λ_{exp} (nm)	λ_{nom} (nm)	Error (%)
1	604	609	0.82
2	568	563	0.89
3	534	539	0.93
4	462	481	3.95
5	444	451	1.55
6	426	434	1.84

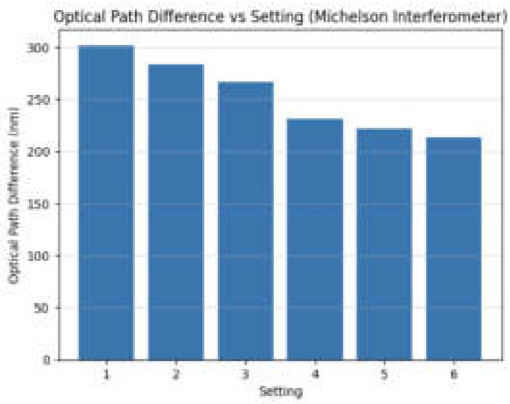

Figure 44. OPD comparison bar chart

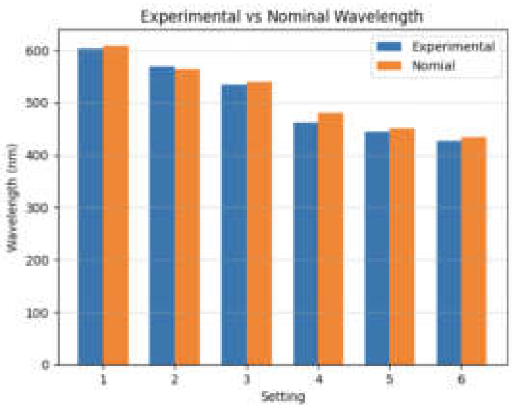

Figure 45. Wavelength comparison (exp vs Nominal)

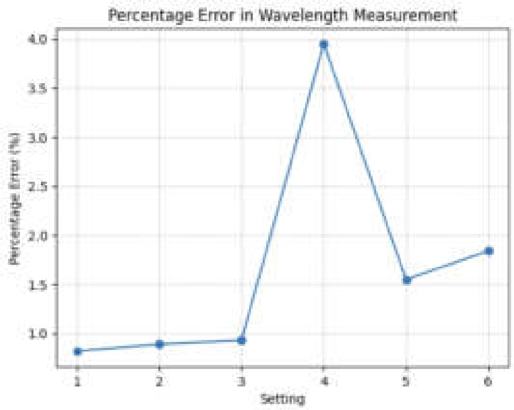

Figure 46. Percentage error in a wavelenght measurement

3.7. Calculation of Relative Velocity for an External Observer in a Simulation[7]

Table 8. Relativistic analysis of left and right sensor times

Point	Left (s)	Right (s)	γ	v (m/s)
1	0.91	1.12	1.015	9.3×10^6
2	0.86	1.21	1.035	1.41×10^7
3	0.77	1.45	1.11	1.99×10^7
4	0.74	1.57	1.155	2.33×10^7
5	0.63	2.44	1.535	2.53×10^8
6	0.58	3.85	2.215	2.64×10^8

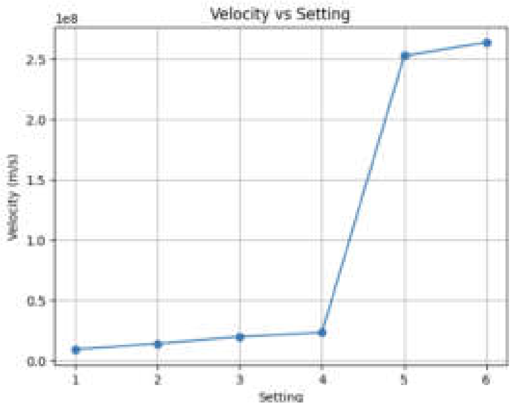

Figure 47. Relativistic Velocity as a Function of Simulation Point)

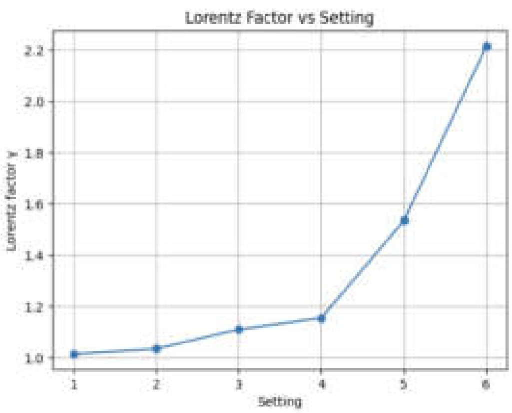

Figure 48. Variation of the Lorentz Factor Across Simulation Points

4. DISCUSSION

4.1. Optical Interference Experiments

Tables 1 and 2 show the settings for the single- and double-slit experiments. The collected data were processed to calculate the angle at which the first minimum occurred for each configuration. The number of bright fringes observed before the first minimum was determined from the graphs and used to obtain the experimental values, from which the percentage errors were calculated.

The comparison between theoretical and experimental values (Figure 31 and Fingure 32) shows that most settings agree well with theory, with relatively small errors. However, setting 2 shows a large deviation, likely due to misalignment, incorrect fringe identification, or sensor limitations. A similar trend is observed in the missing order, where most values match theory except for the same setting. In general, the results confirm the expected behavior, while also showing the sensitivity of the experiment to alignment and measurement accuracy.

The figures 37 and 39 did not record more than three maxima within the diffraction envelope; however, they show a detailed interplay between interference and diffraction, as they exhibit the largest number of bright fringes before the intensity decays, as illustrated in Figure 7 in the first section.

The results for settings 4, 5, and 6 were similar. All three show the same number of maxima (seven, including the central maximum). However, setting 7 shows a significant increase in laser intensity, as every fringe exhibits a much higher intensity compared to the other two configurations.

Figure 43 contrast the results between single and double slit experiments. As theory states, the intensity is given by diffraction, whereas bright fringes is due to interference. Unfortunately, an interference pattern without diffraction is kind of a problem, since both effects are always present together in real situations.

Noise was present during the experiment, and in order to achieve an ideal setting, it is recommended to avoid any external sources of light such as sunlight or lamps. A completely dark environment allows the laser intensity to be reduced and produces a clearer interference pattern.

The sensor should be moved very slowly so that it obtains more data points, producing a smoother graph in the Vernier interface.Figure 39 shows bright spots with larger peaks shifted toward the left half of the graph, indicating a possible misalignment of the optical setup relative to the sensor axis.

4.2. Analysis of Interference Results and Michelson Interferometer Performance

Figure 45 shows the optical path difference for each setting required to transition from one constructive interference pattern to the next. This corresponds to a phase shift of one wavelength, which is the parameter of interest. Therefore, the wavelength is obtained as twice the measured optical path difference (OPD), since the mirror displacement produces a round-trip path change.

The results are presented in Figure 45, where the experimental values are compared with the nominal wavelength of the laser. The corresponding percentage errors are shown in Figure 46. Although most settings show good agreement, setting 4 exhibits a significant deviation. Overall, the trend suggests that shorter wavelengths tend to produce larger measurement errors, likely due to increased sensitivity to alignment and resolution limitations.

The Michelson interferometer simulations were a good starting point to interact with the experiment. The basic concepts were understood, and the two simulations offered different variables to adjust in order to obtain better results.

Next step for future research is to work with real experimental equipment. A wide range of experiments can be performed using a Michelson interferometer in the laboratory at LaGuardia Community College, including measuring the wavelength of a light source, determining the refractive index of air or gases, and estimating thin film thickness, among others [9].

The original interferometer setup used a much larger arm length, whereas the proposed setup is limited to approximately less than one meter, making the experiment less accurate compared to the original design. Additional mirrors could be installed so as to increase the effective arm length before the two light beams recombine at the beam splitter.

4.3. Relativity simulation and proposal

Figure 47 shows the velocity at which the vehicle is moving, which causes the sensors to measure different time intervals. Figure 48 shows the variation of the Lorentz factor as the velocity of the vehicle increases. For small velocities, γ approaches 1, and the motion can be accurately described using Newtonian mechanics. However, as the velocity increases and becomes comparable to the speed of light, the Lorentz factor grows significantly, and the system must be described using Einstein's theory of special relativity.

The simulation was very helpful to understand the concepts of time dilation and the loss of simultaneity. The interface is user-friendly and allows switching between observers, which is something no possible in real life yet. Since this theory originates from a thought experiment proposed by Albert Einstein, the simulation allows us not only to visualize but also to interact with these ideas.

The ISS travels at high speed; however, for time dilation experiments, much higher speeds are required to produce a noticeable effect. Nevertheless, thanks to atomic clocks, it is possible to measure time intervals with very high precision, on the order of nanoseconds or microseconds.

5. CONCLUSIONS

5.1. Optics: Interference and Diffraction

The results obtained from the single- and double-slit experiments showed overall good agreement with the expected theoretical behavior. The measured angles, fringe patterns, and calculated wavelengths were consistent with the predictions, confirming the relationship between interference and diffraction. Although most settings produced small percentage errors, some deviations were observed, particularly in specific configurations, which can be attributed to misalignment, limitations in fringe identification, and sensor resolution.

Throughout the experiment, it became clear that interference and diffraction cannot be treated as completely separate effects. The diffraction envelope controls the overall intensity distribution, while interference determines the position and number of bright fringes within that envelope. This interplay was evident in the collected data and graphical analysis.

A proposal for future experiments in optics is to eliminate any external light sources other than the laser, while also increasing safety measures by using protective glasses. Varying the wavelength of the light for the same experimental setup would improve the accuracy of the results. Additionally, using a white light source could allow the observation of different interference patterns simultaneously, as each wavelength would produce its own pattern on the same screen.

5.2. Michelson Interferometer and Michelson–Morley

The Michelson interferometer proved to be a very powerful tool, as it allows a wide range of experiments to be performed with high precision. From measuring the wavelength of light to determining refractive indices and thin film thickness, the number of possible experiments shows how versatile this instrument is in optics.

The Michelson–Morley experiment, although considered a "failed" experiment at the time, played a fundamental role in physics. Its inability to detect the ether provided strong evidence that led to the development of Einstein's theory of relativity, changing the way space and time are understood.

In modern physics, interferometers have opened a new era of technology. One of the most important applications is the detection of gravitational waves, where large-scale interferometers use the same basic principle to measure extremely small changes in distance. This shows how a classical optical experiment evolved into a key tool for exploring the universe

5.3. Special Relativity

Albert Einstein's special relativity did not wipe out Newton's laws; instead, it extended them. For small velocities, special relativity reduces to classical mechanics, and both give the same results. However, when speeds become comparable to the speed of light, relativistic effects become significant, and the theory introduces limits through the Lorentz factor.

Experimenting with relativity is considered expensive compared to other types of experiments, since it requires very high speeds, highly precise measurements such as atomic clocks, and advanced instrumentation capable of detecting extremely small differences in time and distance. Many of these experiments depend on large-scale facilities such as particle accelerators or space-based platforms, which makes them less accessible for direct laboratory reproduction. The proposed experiment in this paper reflects the complexity of being carried out in order to obtain measurements that test time dilation.

For educational purposes, simulations are a great help in becoming familiar with the fundamental concepts of special relativity. As previously discussed, much of this theory originated from Einstein's thought experiments involving a train. Nowadays, simulations offer more than just mental visualization; they allow us to interact with the system by switching between observers inside and outside the train. In the future, with the help of technologies such as virtual reality, more advanced simulations could make it possible to directly experience effects like length contraction and time dilation while moving at relativistic speeds.

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